

व्यावसायिक परीक्षण रिपोर्ट (प्रारंभिक)

संख्या / No. : POWER WEEDER-221/3299/2025

COMMERCIAL TEST REPORT (INITIAL)

माह / Month : March, 2025

(यह परीक्षण रिपोर्ट 31/03/2032 तक वैध है। / THIS TEST REPORT IS VALID UP TO: 31/03/2032)



**FARMWORLD, FWMT310D,
POWER WEEDER**



भारत सरकार

Government of India

कृषि एवं किसान कल्याण मंत्रालय

Ministry of Agriculture and Farmers Welfare

कृषि एवं किसान कल्याण विभाग

Department of Agriculture and Farmers Welfare

उत्तरी क्षेत्र कृषि मशीनरी प्रशिक्षण एवं परीक्षण संस्थान

Northern Region Farm Machinery Training and Testing Institute

ट्रैक्टर नगर, सिरसा रोड, हिसार, (हरियाणा)- 125 001

Tractor Nagar, Sirsa Road, HISAR (Haryana)-125 001

[ISO 9001:2015 CERTIFIED]

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POWER WEEDER-221/3299/2025	FARMWORLD, FWMT310D,
	POWER WEEDER (COMMERCIAL-INITIAL) [THIS REPORT IS VALID UP TO: 31/03/2032]

Type : Self Propelled, Walk Behind Type

Make : FARMWORLD

Model : FWMT310D

Name and address of manufacturer : Farmworld Agrotech Private Limited
Plot No. 373, Road-U,
Khirasara Industrial Estate, Taluka: Lodhika,
District-Rajkot, Gujarat-360021

Name and address of applicant : Farmworld Agrotech Private Limited
Plot No. 373, Road-U,
Khirasara Industrial Estate, Taluka: Lodhika,
District-Rajkot, Gujarat-360021

Test conducted at : Government of India,
Northern Region Farm Machinery
Training and Testing Institute,
Tractor Nagar, Sirsa Road,
Hisar-125 001 (Haryana)

THIS TEST REPORT VALID UP TO : 31st March, 2032

[vide F. No. 13-22/2020- M&T (I&P) dated 12.12.2023]

Report No. POWER WEEDER – 221/3299/2025

Month : March

Year : 2025



GOVERNMENT OF INDIA
NORTHERN REGION FARM MACHINERY TRAINING & TESTING INSTITUTE
TRACTOR NAGAR, SIRSA ROAD, HISAR-125001 (HARYANA)
[ISO 9001:2015 CERTIFIED]

POWER WEEDER-221/3299/2025	FARMWORLD, FWMT310D,
	POWER WEEDER (COMMERCIAL-INITIAL) [THIS REPORT IS VALID UP TO: 31/03/2032]

Type of Test : **COMMERCIAL - INITIAL**

Period of Test : January, 2025 to February, 2025

Test Report No. : **POWER WEEDER – 221/3299/2025**

Month and Year : **March, 2025**

-
- i) The results reported in this report are observed values and no corrections have been applied for atmospheric and site conditions.
- ii) The data given in this report pertains to the particular machine submitted by the applicant for test.
- iii) The results presented in this report do not in any way attribute to the durability of the machine.
- iv) **This report should not be reproduced in part or full without prior permission of the Director, Northern Region Farm Machinery Training & Testing Institute, Hisar-125001.**
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SELECTED CONVERSIONS

1. **Force**
 - 1 kgf = 9.80665 N
 - = 2.20462 lbf
2. **Power**
 - 1 hp = 1.01387 Metric hp (Ps)
 - = 745.7 W
 - 1 Ps = 735.5 W
3. **Pressure**
 - 1 psi = 6.895 kPa
 - 1 kgf/sq.cm = 98.067 kPa = 735.56 mm of Hg
 - 1 bar = 100 kPa = 10 N/sq.cm.
 - 1 mm of Hg = 1.3333 m-bar
4. **Area**
 - 1 ha = 10000 m² = 2.471 Acres



POWER WEEDER-221/3299/2025	FARMWORLD, FWMT310D, POWER WEEDER (COMMERCIAL-INITIAL)
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IMPORTANT

This machine named FARMWORLD, FWMT310D power weeder was submitted at this institute “to test only with the tilling device attached therewith in the dry land operation, that too, in the broad day light with sufficient visibility”.



POWER WEEDER-221/3299/2025	FARMWORLD, FWMT310D,
	POWER WEEDER (COMMERCIAL-INITIAL) [THIS REPORT IS VALID UP TO: 31/03/2032]

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POWER WEEDER-221/3299/2025	FARMWORLD, FWMT310D,
	POWER WEEDER (COMMERCIAL-INITIAL) [THIS REPORT IS VALID UP TO: 31/03/2032]

1. SCOPE OF TEST

The scope of the test was limited to check and assess the following :

- 1.1 Specifications and other data furnished by the applicant.
- 1.2 Engine performance
- 1.3 Mechanical vibration measurement
- 1.4 Noise level measurement
- 1.5 Air cleaner oil pull-over test
- 1.6 Field performance with rotary tiller
- 1.7 Operator's comfort and safety
- 1.8 Ease of operation and handling
- 1.9 Breakdowns and repairs
- 1.10 Wear of critical components
- 1.11 Hardness & chemical analysis of rotor blades

2. METHOD OF SELECTION

- i) Test requested by : The applicant
- ii) Method of selection : Randomly selected by the representative of the Institute through virtual mode.
- iii) Serial numbers of the available power weeder at the time of Random selection of the test sample : The test sample was selected out of five number of samples, made available by the applicant for random sampling.

Sr. No.	Sr. No. of machine
1.	GJ/6774/3538/2024/0281
2.	GJ/6774/3538/2024/0250
3.	GJ/6774/3538/2024/0267
4.	GJ/6774/3538/2024/0264
5.	GJ/6774/3538/2024/0268

- iv) Serial number of sample selected : **GJ/6774/3538/2024/0250**

3. TEST CODE/PROCEDURE

IS:4905-2015	:	Methods for Random Sampling and Randomization procedure
IS:15925-2012 (Reaffirmed 2020)	:	Walk – behind powered rotary tillers – Definitions, Safety Requirements and Test Procedures
IS:9935-2002 (Reaffirmed 2022)	:	Power tiller – Test code
IS:9980-1988 (Reaffirmed 2019)	:	Guidelines for Field Performance and Haulage Tests of Power Tillers
IS:12036-1995 (Reaffirmed 2019)	:	Agricultural Tractors-Test Procedures – Power Tests for Power Take-off
IS:6690-1981 (Reaffirmed 2022)	:	Specification for Blades for Rotavator for Power Tillers (First revision)
IS: 13539-2018	:	Power Tillers – Recommendations on Selected Performance and Other Characteristics (Second Revision)
Critical Technical Specifications	:	Critical technical specification issued by Ministry, vide letter No. 13-9/2019 - M&T (I&P) - Part dated 26.04.2019 and F. No. 9-1/2019 M&T (I&P) dated 20.08.2019.

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[THIS REPORT IS VALID UP TO: 31/03/2032]	

4. SPECIFICATIONS

4.1 General

Type of machine : Self Propelled, Walk Behind Type
 Make : FARMWORLD
 Model : FWMT310D
 Serial No. : GJ/6774/3538/2024/0250
 Name and address of manufacturer : Farmworld Agrotech Private Limited
 Plot No. 373, Road-U,
 Khirasara Industrial Estate, Taluka: Lodhika,
 District-Rajkot, Gujarat-360021

Name and address of applicant : Farmworld Agrotech Private Limited
 Plot No. 373, Road-U,
 Khirasara Industrial Estate, Taluka: Lodhika,
 District-Rajkot, Gujarat-360021

Year of manufacture : 2024
 Country of origin : India

4.2 Details of prime mover

Name and address of manufacturer : Farmworld Agrotech Private Limited
 Plot No. 373, Road-U,
 Khirasara Industrial Estate, Taluka: Lodhika,
 District-Rajkot, Gujarat-360021

Type : Four stroke, air cooled, single cylinder, diesel engine
 Make : FARMWORLD
 Model (apa) : 178/FWMT310D
 Serial No. : GJ/6774/3538/2024/0249
 Country of origin : India
 Year of manufacture : 2024
 Engine speed (recommended setting), rpm (apa) :

High idle speed : 3888±100

Low idle speed : 1300±100

Rated speed, rpm : 3600
 Speed at maximum torque, rpm : 2880
 Rated engine speed for field, rpm : 3600

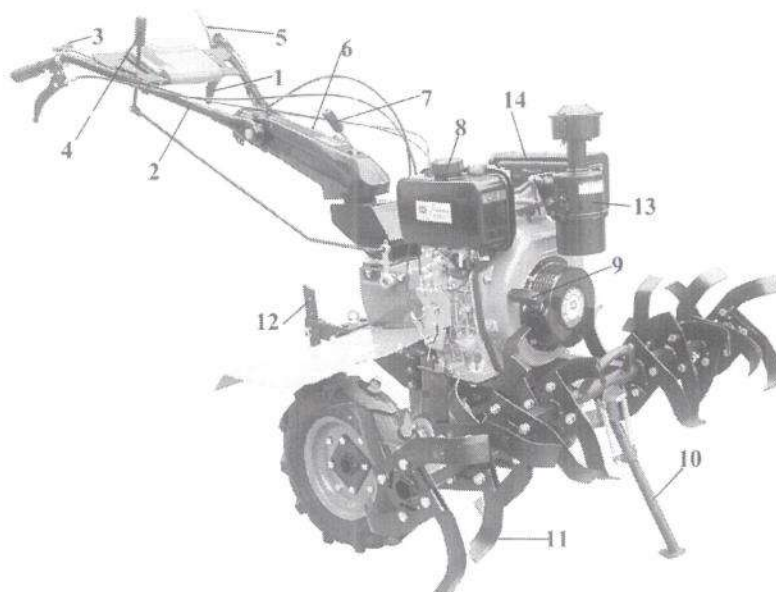
4.3 Cylinder & cylinder head

Number : One
 Disposition : Inclined
 Bore/Stroke, mm (apa) : 78/62
 Capacity, cc (apa) : 296
 Type of valve : Over head



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	POWER WEEDER (COMMERCIAL-INITIAL) [THIS REPORT IS VALID UP TO: 31/03/2032]

	Valve clearance, mm (apa)	Inlet : 0.15 Exhaust : 0.15
	Compression ratio (apa)	: 20:1
4.4	Fuel supply system	
	Type of fuel feed	: Gravity feed
4.4.1	Fuel tank	
	Material	: Sheet metal
	Capacity of fuel tank, l	: 3.0
	Location of fuel tank	: On top of the engine
	Provision for draining of sediments/ water	: Not provided
	Fuel filter	: Provided in fuel tank
	Fuel on/off knob	: Provided
4.4.2	Fuel injection pump	
	Make	: Not specified
	Model (apa)	: 178F
	Type	: Plunger
	Method of drive	: Through cam shaft
4.4.3	Fuel injector	
	Make	: Not specified
	Model (apa)	: 178F nozzle
	Type	: Multi hole
	Serial no./Part no.	: Not specified
	Injection pressure, kg (apa)	: 224 to 235
4.4.4	Governor	
	Make	: Not specified
	Model	: Not specified
	Type	: Centrifugal, variable speed type
4.5	Air intake system	
4.5.1	Pre-cleaner	: Not provided
4.5.2	Air cleaner	
	Type	: Oil bath type
	Make	: Not specified
	Model	: Not specified
	Location	: In front of engine
	Recommended service schedule (apa)	: After every 50 hours of operation.
	Recommended grade of oil (apa)	: SAE 10W30
4.6	Exhaust	
	Type of silencer	: Rectangular
	Location of silencer	: At LHS of the engine.
	Spark arresting device, if any	: Not provided

**Key Words:**

- | | |
|---------------------------------|----------------------------------|
| 1. Reverse clutch lever | 2. Handle bar |
| 3. Accelerator lever | 4. Main gear shifting lever |
| 5. Tilling clutch lever | 6. Handle height adjusting lever |
| 7. Handle angle adjusting lever | 8. Fuel tank cap |
| 9. Recoil starter | 10. Stand |
| 11. Rotor blade | 12. Depth adjusting lever |
| 13. Air cleaner | 14. Exhaust |

FIG.1: FARMWORLD, FWMT310D, POWER WEEDER**4.7 Lubrication system**

Type	: Splash
Oil filter	: Not provided
Oil capacity, l	: 0.6
Recommended grade of lubricant	: SAE 10W30
oil (apa)	
Oil change period (apa)	: First change after 20 hours of operation and subsequently after every 100 hours of operation .

4.8 Cooling system

Type	: Forced air cooling
Dia. of blower, mm	: 232
No. of vanes	: 24

4.9 Starting system

Type	: Manual, recoil starting
Ignition system	: Compression ignition
Any other provision for easy starting	: Decompression lever is provided



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	POWER WEEDER (COMMERCIAL-INITIAL) [THIS REPORT IS VALID UP TO: 31/03/2032]

	Aid for cold starting	: Not provided
4.10	Generator	
	Make	: Not specified
	Model	: Not specified
	Type	: Charge coil
	Method of drive	: Through crankshaft
4.11	Power transmission system	
	Type	: Engine power is transmitted to rotor through primary main gear box & secondary reduction gear box unit. The primary gear box consists of spur gear combination, whereas secondary reduction has bevel & pinion combination.
4.11.1	Clutch	
	Make	: BST
	Type of clutch	: Multi friction plate, wet type
	No. of friction plate	: 5.0
	Dia. of friction plate, mm	: 120
	Thickness, mm	: 2.80
	Method of operation	: One hand lever is provided at LHS handle bar
	Location of clutch	: In between engine and transmission system
4.11.2	Transmission system	
	Type	: Sliding mesh gear box
	Make	: FW
	No. of speed	: 03 (02 forward & 01 reverse)
	Recommended grade of oil (apa)	: SAE 90
	Oil capacity, l (apa)	: 2.0
	Oil change period (apa)	: After every 50 hours of operation.
4.11.3	Rotary gear box	
	Type	: Bevel and pinion
	No. of teeth on drive pinion	: 10
	No. of teeth on driven crown	: 43
	Reduction ratio	: 4.3:1
	No. of teeth on drive pinion	: 10
	No. of teeth on driven pinion	: 36
	Reduction ratio	: 3.6:1
	Oil capacity, l (apa)	: 2.0 (Common with main gear box)
	No. of engine revolution in one revolution of drive wheel	
	Gear-1	: 51.2
	Gear-2	: 29.4
	Reverse gear	: 68.3



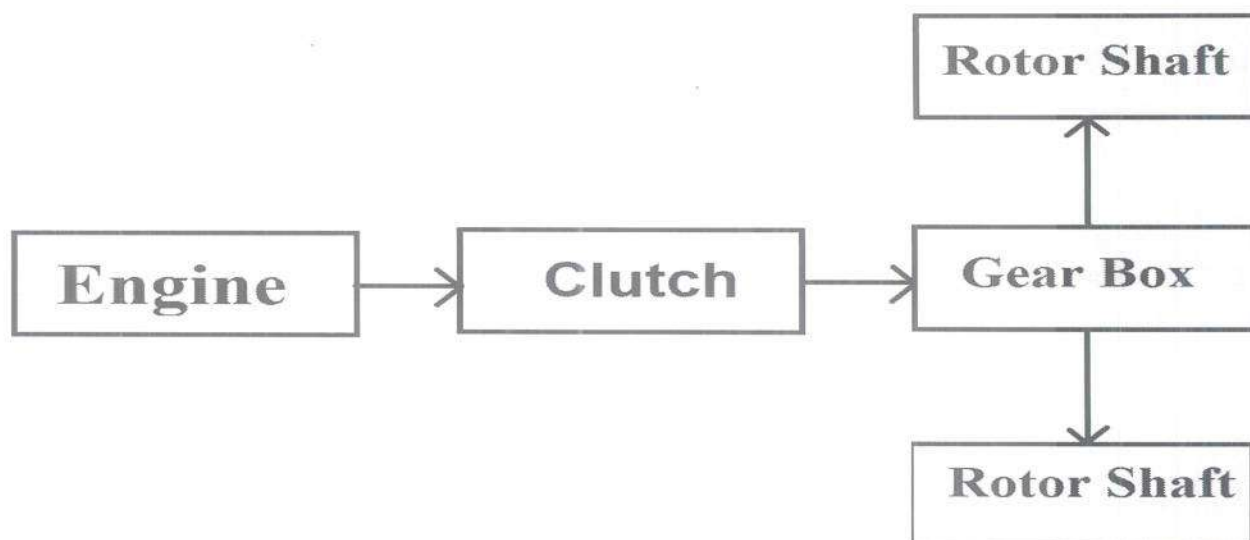


FIG. 2 : SCHEMATIC POWER TRANSMISSION DIAGRAM

4.12 Rotor**4.12.1 Rotor cover**

Material : MS sheet

Size, mm:

Length : 350

Width : 245 to 405 (extension)

Thickness : 1.3

Method of fixing : Bolted to main frame

4.12.2 Rotor shaft

Material (apa) : High carbon steel (SAE 1045)

Type of rotor axle : Hexagonal

Width across flat of rotor axle, mm : 31.7

Length of shaft, mm : 400

Length of extension shaft, mm : 145

No. of flanges : 8 (4+4)

Type of flanges : Square

Size of flanges, mm : 95 × 95

Thickness of flange, mm : 4

Distance between two flanges, mm : 136

No. of blades on each flange : 4

Overall length of rotor, mm : 1130

Diameter of rotor with blades, mm : 355

Method of fixing of blade : By nuts & bolts

No. of disc : None

4.12.3 Rotor blades

Number : 32

Type : 'J' shape

Marking (if any) : RONG



Thickness, mm	: 4.50
Width of beveled edge, mm	: 7.16 to 12.30
No. and dia. of hole on each blade for fixing it to the flanges, mm	: 02 and 11.00
Arrangement of blades on the axle	: LHS & RHS mounted alternatively.
Material of blade	: High carbon steel

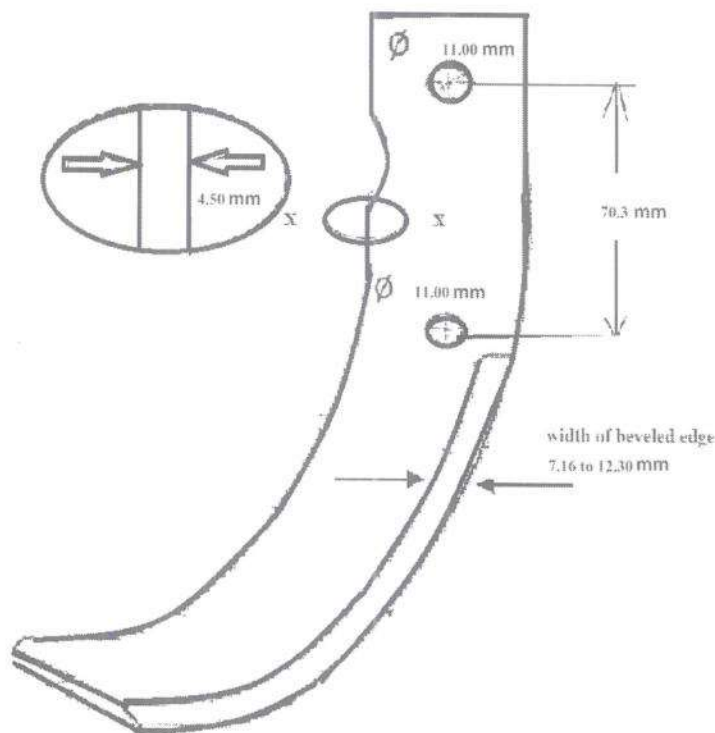


FIG. 3 : ROTOR BLADE

4.13 Depth control mechanism

Type	: M.S. flat
No. of rod	: 01
Material (apa)	: Mild steel
Size of flat, mm :	
Length	: 435
Width	: 37
Thickness	: 11
Provision for depth adjustment	: 8 holes of size 11 mm diameter at distance of 40 mm are provided on M.S. flat for depth adjustment.

4.14 Steering handle bar

Material (apa)	: M.S. pipe
Dia. of pipe, mm	: 25.2
No. of hand grips	: Two
Length of grip, mm	: 116
Material of grip (apa)	: Rubber



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	Dia. of grip, mm	:	37	
	Provision for handle height adjustment	:	Provided	
	Height of handle bar from ground level, mm	:	Maximum : 1226 Minimum : 765	
	Provision for angle adjustment	:	Provided	
4.15	Stand	:	Provided	
4.16	Transport wheel			
	Type	:	Pneumatic	
	Make	:	RX	
	Number	:	02	
	Size, mm	:	4.00 – 8, 4 PR	
	Recommended pressure at 300 kg load, kPa	:	330	
4.17	Controls			
	At LHS			
	i) Tilling clutch lever			
	ii) Handle height adjusting lever			
	iii) Emergency stop lever			
	At RHS			
	i) Accelerator lever			
	ii) Reverse lever			
	iv) Main gear shifting lever			
	At Centre			
	i) Handle angle adjustment lever			
4.18	Overall dimensions, mm		Field working condition	Transport
	Length	:	2100	2100
	Width	:	1130	965
	Height	:	1000	1000
	Mass, Kg	:	118	
4.19	Color of machine			
	Engine	:	Silver	
	Main drive cover	:	Green	
	Transmission system	:	Green	
	Rotary drive cover	:	Green	
	Fuel tank	:	Black	
	Handle bar	:	Black	



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4.20 Labeling /Identification plate : A labeling plate is riveted on the machine with following information.

FW Honour Your Trust Power Weeder	
Model	: FWMT310D
Engine	: 4 Stroke Diesel
Rated Power	: 4.04 kW (5.4 hp)/3600 rpm
Displacement	: 296
S.F.C	: 290 g/kWh
Tilling Depth	: 100 – 250 mm
Tilling Width	: 750 – 1050 mm
Sr. No.	: GJ/6774/3538/2024/0250
Engine No.	: GJ/6774/3538/2024/0249
Year	: 2024
Farmworld Agrotech Private Limited Plot No. 373, Khirasara Industrial Estate, Rajkot, Gujarat, India www.farmworld.in	

5. FUEL & LUBRICANTS			
5.1	Fuel	: Diesel	
5.2	LUBRICANTS		
Sr. No.	Particulars	As recommended by the manufacturer	As used during the test
1.	Engine sump	SAE 10W30	Oil originally filled in the machine was not changed.
2.	Gear box	SAE 90	

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	POWER WEEDER (COMMERCIAL-INITIAL)
[THIS REPORT IS VALID UP TO: 31/03/2032]	

6. ENGINE RATING TEST AND FUEL CONSUMPTION TEST

Date of test : 04.02.2025
 Model of dynamometer : AG-10
 Dynamometer constant : 9546.305

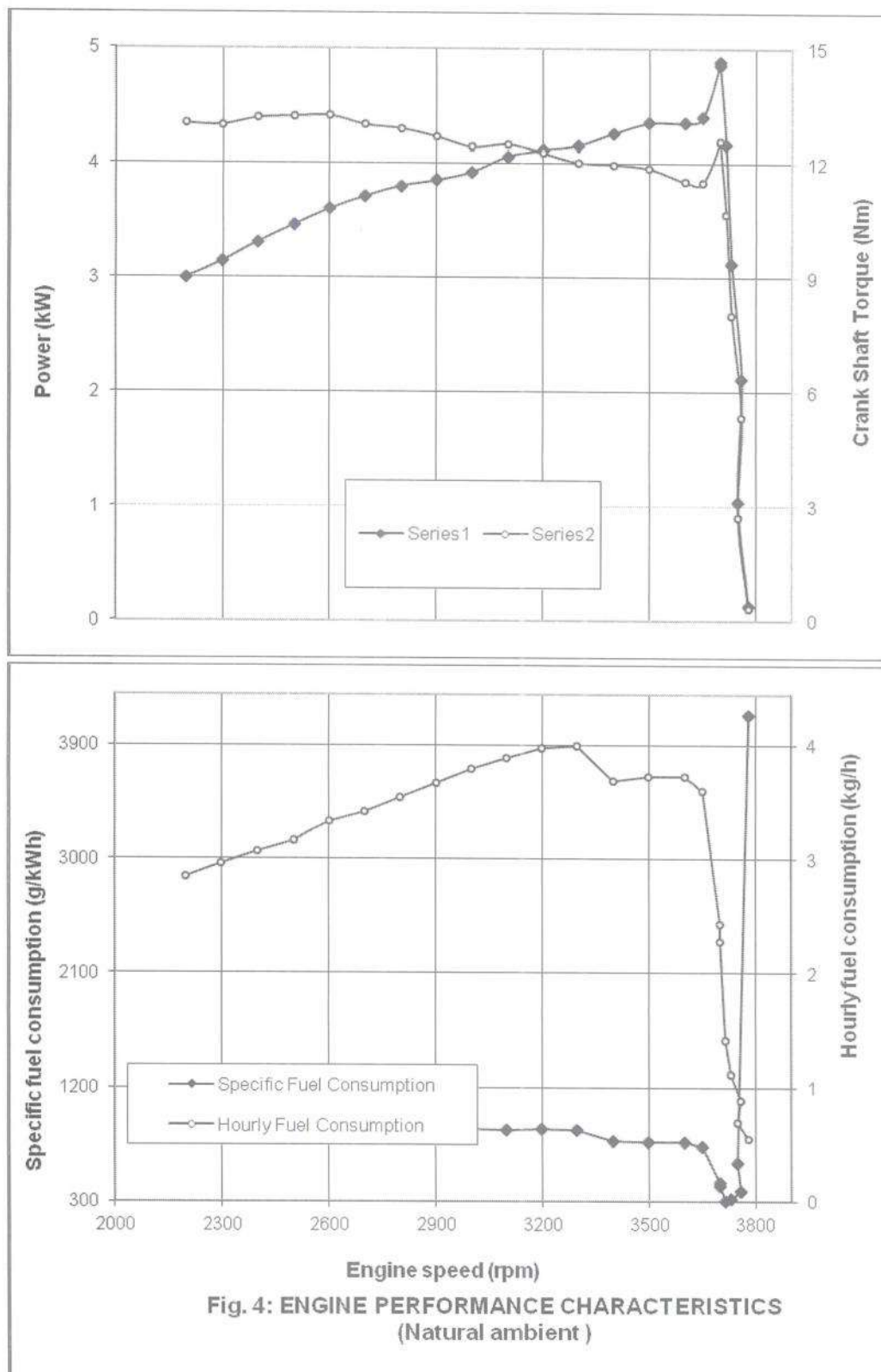
TABLE:1 ENGINE PERFORMANCE TEST

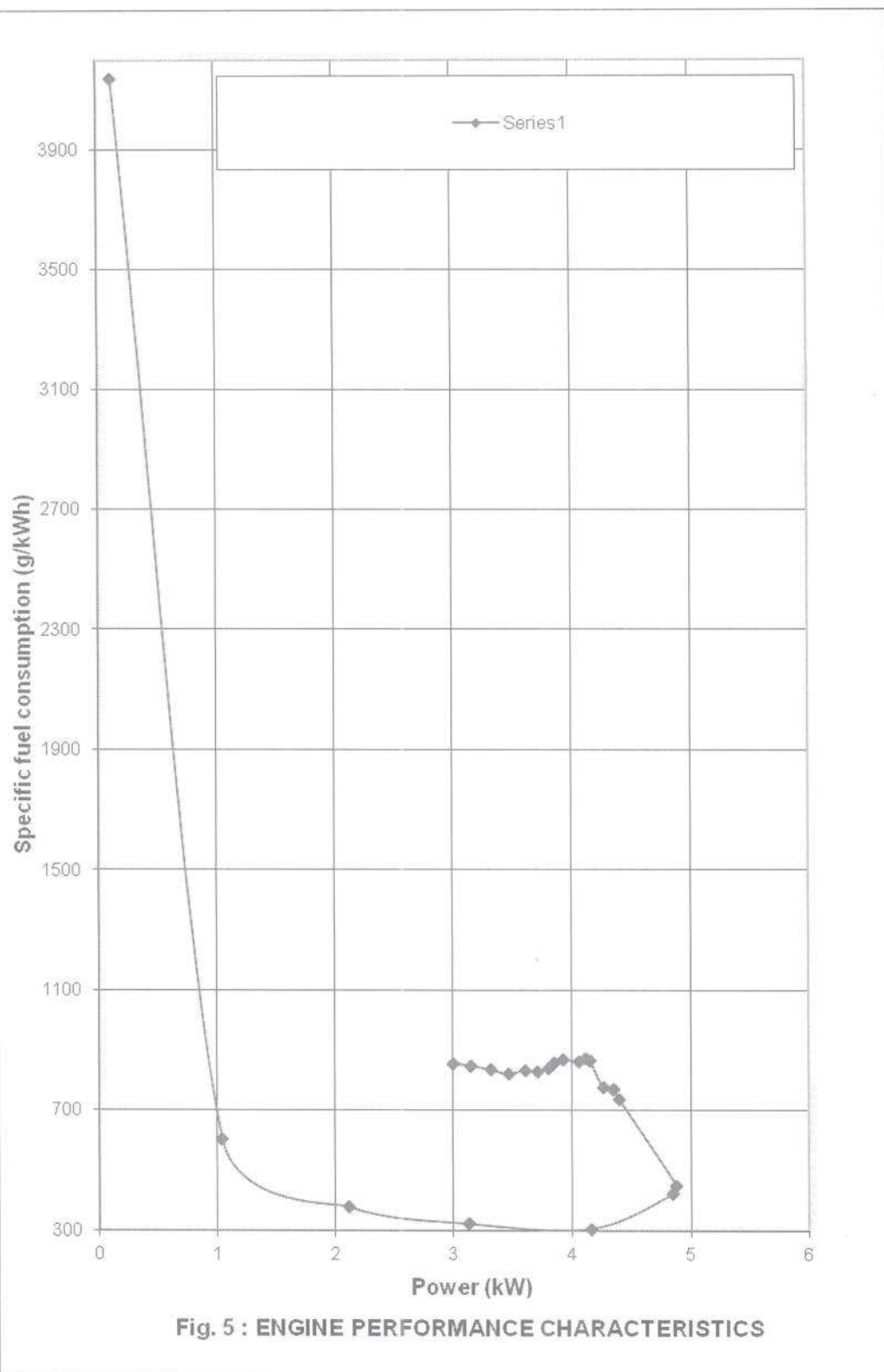
Brake power, kW	Engine speed, rpm	Fuel consumption			Specific energy,
		l/h	kg/h	Specific, gm/ kWh	kWh/l
a) Maximum power 2 hours test					
4.88	3700	2.60	2.19	449.86	1.88
b) Power at rated engine speed: (3600 rpm)					
4.35	3602	3.98	3.36	771.20	1.09
c) Varying load test:					
i) Torque corresponding to maximum power					
4.85	3699	2.43	2.05	421.65	2.00
ii) 85% of torque obtained at maximum power:					
4.16	3716	1.51	1.27	305.66	2.76
iii) 75% of torque defined in (ii)					
3.12	3732	1.19	1.01	322.24	2.61
iv) 50% of torque defined in (ii)					
2.11	3760	0.95	0.80	377.84	2.23
v) 25% of torque defined in (ii)					
1.04	3749	0.75	0.63	603.56	1.40
vi) Unloaded					
0.12	3780	0.59	0.50	4138.33	0.20

TABLE:2 Engine performance parameters

Parameters		Natural ambient
No load engine speed, rpm	:	3780
Crankshaft torque at max. power, Nm	:	12.57
Max. crankshaft torque, Nm	:	13.23
Engine speed at max. crankshaft torque, rpm	:	2601
Range of atmospheric conditions:		
Temperature, °C	:	19.8 to 21.4
Pressure, kPa	:	98.9 to 99.1
Relative humidity, %	:	53.0 to 62.0
Max. temperature, °C		
Engine oil	:	100.5
Fuel	:	25.80
Air intake	:	24.40
Exhaust gas	:	255
Smoke density at 80% of max. power (Bosch No.)	:	0.57







POWER WEEDER-221/3299/2025	FARMWORLD, FWMT310D,
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7. MECHANICAL VIBRATION TEST

The amplitude of mechanical vibration on various assemblies/components of rotary power weeder was recorded by running the machine under stationary condition with accelerator lever recommended for rated engine speed with rotary shaft in running condition. The amplitude of vibration was measured in horizontal and vertical position of the accelerometer and the results are represented in Table -1.

Table 1: Vibration Measurement

Sr. No.	Location	Horizontal Direction (x) (μ)	Vertical Direction (y) (μ)
1.	Steering handle		
	Left	272*	340*
	Right	280*	316*
2.	Throttle lever	260*	380*
3.	Clutch lever	330*	380*
4.	Handle up down adjusting lever	270*	290*
5.	Fender assembly right		
	Left	300*	340*
	Right	280*	310*
6.	Air cleaner	480*	520*
7.	Fuel tank	310*	430*
8.	Reverse lever	260*	220*
9.	Stop lever	280*	301*
10.	Depth control lever	260*	510*
11.	Gear shifting lever	340*	230*
12.	Steering angle adjustment lever	238*	312*

Remarks: The amplitude of mechanical vibration on the various assemblies and sub assemblies as marked (*) may be considered to be on higher side.

8. NOISE LEVEL MEASUREMENT

8.1 Noise at By-stander's position

Date of test	: 24.01.2025
Type of track	: Concrete surface
Background noise level, dB (A)	: 52.7
Atmospheric conditions:	
Temperature, °C	: 23.3
Pressure, mm/hg	: 743.0
Relative humidity, %	: 41.6
Wind velocity, m/sec	: 1.6 to 2.5



TEST DATA:

Sr. No.	Gear Used	Travelling speed before acceleration (kmph)	Noise level, dB (A)	
			Silencer facing microphone	Silencer facing away from microphone
1.	1 st	5.12	79.5	78.6
2.	2 nd	8.93	79.6	78.8

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8.2 Noise at operator's ear level

Date of test : 24.01.2025
Type of track : Soil surface
Background noise level dB (A) : 52.7

Atmospheric conditions

Temperature, °C : 23.3
Pressure, mm/hg : 743.0
Relative humidity, % : 41.6
Wind velocity, m/sec : 1.6 to 2.5
Observed Noise level, dB (A) : 93.3

9. AIR CLEANER OIL PULL OVER TEST

Date : 24.01.2025

Range of atmospheric condition

Temperature, °C : 20.2 to 21.4
Relative humidity, % : 53.2 to 54.2
Pressure, mm/hg : 745.0 to 745.4
Mass of oil in the air cleaner assembly : 52.5
when filled with recommended grade of
oil 5% in excess than mark level, g

Sr. No.	Position	Slope (°)	Loss of oil (g)	Oil pull over (%)	Remarks if any
1.	Horizontal	0°	0.3	0.11	None
2.	Tilt longitudinally with front end up	15°	0.2	0.07	
3.	Tilt longitudinally with rear up	15°	0.3	0.11	
4.	Tilt laterally with right side up	15°	0.4	0.14	
5.	Tilt laterally with left side end up	15°	0.4	0.14	



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10. HARDNESS & CHEMICAL COMPOSITION OF BLADES

Hardness & chemical analysis of primary element of the blades were carried out as per IS: 6690 -1981(Reaffirmed 2022). The details of same are given in Table 2 & 3.

10.1 Table 2 : Hardness of blades

	Requirement as per IS: 6690-1981 (Reaffirmed 2022) (HRC)	Hardness (HRC) as observed	Remarks
At edge portion	56±3	45.9	Does not conform
At shank portion	37 to 45	40.8	Conforms

10.2 Table 3 : Chemical analysis of rotary blade

Constituents	As per IS: 6690-1981 (Reaffirmed 2022)		Composition as observed	Remarks
	Carbon Steel	Silicon Manganese steel		
Carbon (C)	0.70 -0.85	0.50-0.60	0.84	Conforms
Silicon (Si)	0.10 -0.40	1.50-2.00	0.56	Does not conform
Manganese (Mn)	0.50 -1.00	0.50-1.00	0.83	Conforms
Sulphur (S)	0.05 (Max.)	0.05 (Max.)	0.02	Conforms
Phosphorous (P)	0.05 (Max.)	0.05 (Max.)	0.00	Conforms

11. RUNNING - IN

The power weeder was run-in for 0.58 hour before the field performance test as recommended by the applicant. All the fasteners were checked and tightened thereafter.

12. FIELD TEST

The field tests under dry land condition were conducted for 26.56 hours. The field tests were conducted at the recommended field rpm 3600 rpm. In all, 4 tests trials were conducted in sandy loam soil at NRFMTTI Farm, Hisar. The summary of the field test for dry land operation is given in Table-4.

Crop parameters

- | | | |
|-----|--------------------|------------------|
| i) | Type of weed | - Seasonal weeds |
| ii) | Height of weed, cm | - 9.0 to 20.0 |



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Table 4: SUMMARY OF FIELD PERFORMANCE TEST

Sr. No.	Parameter		Range
i)	Type of soil	:	Sandy loam
ii)	Soil moisture, %	:	9.7 to 10.6
iii)	Bulk density of soil, g/cc	:	1.15 to 1.26
iv)	Speed of operation, kmph	:	1.71 to 1.77
v)	Depth of cut, cm	:	5.13 to 5.67
vi)	Width of cut, m	:	1.08 to 1.09
vii)	Area covered, ha/h	:	0.162 to 0.169
viii)	Time required for one ha, h	:	5.92 to 6.17
ix)	Fuel consumption		
		l/h :	0.75 to 0.83
		l/ha :	4.60 to 4.90
x)	Weeding efficiency, %	:	88.86 to 93.08
xi)	Field efficiency, %	:	86.39 to 89.42

13. ADJUSTMENT, DEFECTS, BREAKDOWNS & REPAIR

No noticeable defect/breakdown was observed during test.

14. COMPONENTS/ASSEMBLY INSPECTION AND ASSESSMENT OF WEAR

14.1	Engine :					
	The Engine and other assemblies were dismantled after 37.57 hours of engine operation.					
14.1.1	Cylinder :					
Cylinder bore diameter, mm						
Top Position		Middle position		Bottom Position		Max. permissible wear limit
Thrust	Non-thrust	Thrust	Non-thrust	Thrust	Non-thrust	
78.01	78.00	78.01	78.00	78.01	78.00	78.18
14.1.2	Piston:					
Piston diameter, mm						
Top position		At skirt			Max. permissible wear limit, mm	
Thrust side	Non-thrust side	Thrust side	Non-thrust side	Piston to cylinder clearance, mm	Piston dia. at skirt	Piston to cylinder clearance
77.735	77.740	77.880	Not measured due to piston design constraint	0.13	--	0.48

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14.1.3		Piston rings end gap:					
Ring No.		Ring end gap, mm			Max. permissible wear limit, mm		
		At top	At middle	At bottom			
1 st compression ring		0.25	0.30	0.25	1.00		
2 nd compression ring		0.30	0.30	0.25	1.00		
Oil ring		Not measured due to ring design constraint					
14.1.4		Big end bearing:					
Dia. of crank pin, mm		Dia. of bearing, mm	Clearance, mm		Max. permissible wear limit, mm		
			Diametrical	Axial	Diametrical	Axial	
35.350		35.310	0.040	0.30	0.045	0.40	
14.1.5		Main bearing of crank shaft:					
Sr. No.	Dia. of main journal, mm	Dia. of main bearing, mm	Diametrical clearance of main bearing	End float of crank shaft	Max. permissible wear limit, mm		
					Diametrical	End float of crank shaft	
Ball bearing's are provided at both side hence not applicable							
14.1.6		Piston rings groove clearance:					
Ring No.		Ring groove clearance, mm		Max. permissible wear limit, mm			
1 st compression ring		0.065		0.15			
2 nd compression ring		0.040		0.15			
Oil ring		0.030		0.15			
14.1.7		Valve guide clearance:					
Valve guide diameter, mm		Valve stem diameter, mm		Valve guide clearance, mm		Max. permissible wear limit, mm	
Inlet	Exhaust	Inlet	Exhaust	Inlet	Exhaust	Inlet	Exhaust
5.970	5.965	5.960	5.955	0.01	0.01	0.2	0.2

14.2 Valve guides and valve springs

Valve spring stiffness, N/mm :

Discard limit

Inlet valve : 11.51

8.0

Exhaust valve : 11.00

8.0

14.3 Timing gears

: No noticeable defect was observed.

14.4 Transmission

: No noticeable defect was observed.

14.5 Rotary drive unit

: No noticeable defect was observed.

14.6 Wear of blades:

14.6.1 Mass basis:

The wear of the rotary weeder blades was measured after 27.14 hours of field operation and the observations are as under:



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Sr. No.	Initial mass, g	Mass after 27.14 hours g	Loss of mass, g	Percent wear, %	Percent wear per hour, %
1.	298.0	294.3	3.7	1.24	0.05
2.	307.5	304.1	3.4	1.11	0.04
3.	308.5	304.9	3.6	1.17	0.04
4.	306.0	300.2	5.8	1.89	0.07
5.	307.9	303.4	4.5	1.46	0.05
6.	299.8	296.5	3.3	1.10	0.04
7.	312.8	308.6	4.2	1.34	0.05
8.	307.2	302.6	4.6	1.50	0.06

15. CRITICAL TECHNICAL SPECIFICATIONS

(Vide Ministry's communication F. No. 9-1/2019-M&T (I&P) Dated 20.08.2019)

Sr. No.	Parameters	Specifications	Observed	Remarks
1.	Type	Self-propelled, walk behind	Self propelled, walk behind type	Conforms
2.	Working width, mm	300 - 1500	1130	Conforms
3.	Type of engine	Compression/Spark ignition	Compression ignition	Conforms
4.	Starting method	Manual/recoil/self-starting	Recoil	Conforms
5.	Type of clutch	Dry/Wet	Wet	Conforms
6.	Type of primary gear box	Sliding/Constant mesh or combination of both	Sliding mesh	Conforms
7.	Type of secondary gear box	Gear type, chain & sprocket type	Gear type	Conforms
8.	Material for rotor shaft	SAE 1045 (CRS) / EN8 / EN9	High carbon steel (SAE 1045)	Conforms
9.	No. of flanges	4 - 10	08	Conforms
10.	Types of flanges	Square/circular/rectangular	Square	Conforms
11.	Distance between consecutive flanges, mm	80 to 150	136	Conforms
12.	No. of blades in each flange	3 - 6	04	Conforms

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13.	No. of rotor blade	12 (Min.)	32	Conforms
14.	Thickness of rotor blade, mm	5 (Min.)	4.50	Does not conform
15.	Material of blade	Boron (28MnCrB5) / High carbon steel of Grade EN42J	High carbon steel	Conforms
16.	Hardness of blade, HRC	38 (Min.)	45.9	Conforms
17.	Shape of rotor blade	C / J shape	J shape	Conforms
18.	Provision for handle height adjustment	Must be provided	Provided	Conforms
19.	Provision for handle rotation	Optional	Provided	Conforms
20.	Provision for emergency stop of engine	Must be provided	Provided	Conforms
21.	Provision for easy start of engine	Must be provided	Provided	Conforms
22.	Provision for shield/cover to prevent flying of mud & stone from rotor	Must be provided	Provided	Conforms
23.	Depth control mechanism	Must be provided	Provided	Conforms
24.	Provision for transport wheels	Optional	Provided	Conforms
25.	Provision for cover on exhaust	Must be provided	Provided	Conforms
26.	Direction of exhaust emission away from operator	Must be provided	Provided	Conforms
27.	Marking/labeling machine	The labeling plate should be riveted on the body of machine having Name and address of manufacturer & Applicant, Country of origin, Make, Model, Year of manufacturer, Serial number, Engine number, Engine HP, rated rpm & SFC.	Partially meet the requirement	Partially conform
28.	Literature	Operator manual, service manual and Parts catalogue should be provided.	Provided	Conforms



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16. COMMENTS & RECOMMENDATIONS

- 16.1**
- The maximum power is observed as 4.88 kW (6.54 hp) against of declared maximum power 4.4 kW (5.90 hp).
 - The specific fuel consumption is observed as 450 g/kWh against the declaration of 290 g/kWh which is on higher side and should be looked into for corrective action.

16.2 Mechanical vibration

The amplitude of mechanical vibration marked as (*) in the relevant chapter, are on higher side. It is not just directly concerned with operator's health, safety and comfort, but also adversely affects the useful life of the components. In view of above, this deserved to be given top priority for necessary corrective action.

- 16.3** The chemical composition of blades does not conform to the requirements of IS: 6690-1981 (Reaffirmed 2022). This needs to be looked into for necessary corrective action.
- 16.4** The hardness of blades does not conform to the requirements of IS: 6690-1981 (Reaffirmed 2022). This needs to be looked into for necessary corrective action.

The thickness of blade do not meet the requirement of Critical Technical Specification issued by Ministry.

16.5 Field performance test

No noticeable defect was observed during field test.

A. Rate of work

- The speed of operation ranged from 1.71 to 1.77 kmph during the test. The rate of work varied from 0.162 to 0.169 ha/h.
- The fuel consumption varied from 0.75 to 0.83 l/h.
- The fuel consumption per unit area of operation varied from 4.60 to 4.90 l/ha.

B. Quality of work

- Weeding efficiency of machine ranged from 88.86 to 93.08 % and the field efficiency ranged from 86.39 to 89.42 %



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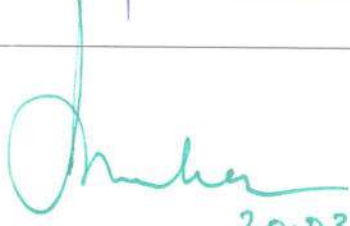
17. TECHNICAL LITERATURE

The following literatures were provided by the applicant.

- Owners manual
- Parts catalogue
- Service manual

However, operators manual needs to be updated as per IS: 8132-2023.

TESTING AUTHORITY

Er. SANJAY KUMAR AGRICULTURAL ENGINEER	
Dr. MUKESH JAIN DIRECTOR	 20.03.2025

18. APPLICANT'S COMMENT

Para No.	Our reference	Applicant comments
18.1	16.1 (ii)	We will work on comments for better result of fuel consumption, In support with our R & D and production team. We will make all possible necessary changes in our product future.
18.2	16.2	We will work on it and take necessary steps, in support with our R & D and production team. We will provide anti vibration rubber grips at handle bars for operator comfort, safety and health. We will also look into adding vibration damping gaskets at all joints wherever possible. We make all possible necessary changes in our product in future.
18.3	16.3	We will provide as per the IS. We will make all possible necessary changes in our product in future.
18.4	16.4	We will provide as per the IS. We will make all possible necessary changes in our product in future.



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ANNEXURE – I

FIELD DATA SHEET OF ROTARY POWER WEEDER

Place: NRFMTTI Farm, Hisar
Type of soil: Sandy loam

Test No.	Date	Duration of test	Speed of operation, kmph	Avg. width of cut, m	Avg. depth of cut, cm	Avg. soil moisture, %	Bulk density, g/cc	Fuel consumption		Area covered, ha/h	Time required for one hectare, h/ha	Field efficiency, %	Weeding efficiency, %
								l/h	l/ha				
1	2	3	4	5	6	7	8	9	10	11	12	13	14
1.	16.01.2025	2.58	NR	NR	-	-	-	-	-	-	-	-	-
2.	17.01.2025	6.17	1.75	1.09	5.67	10.0	1.26	0.77	4.65	0.165	6.06	86.39	89.47
3.	20.01.2025	6.39	1.77	1.08	5.42	10.6	1.20	0.79	4.70	0.169	5.92	87.56	93.08
4.	21.01.2025	6.37	1.75	1.08	5.21	9.8	1.15	0.83	4.90	0.169	5.92	89.42	88.86
5.	22.01.2025	5.05	1.71	1.08	5.13	9.7	1.15	0.75	4.60	0.162	6.17	87.57	92.95

